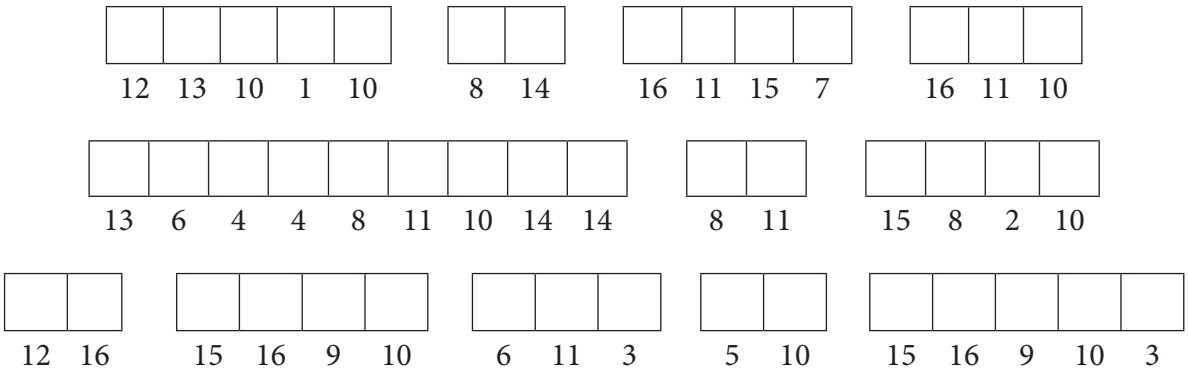
WORKSHEET

Permutations



George Sand

Questions

How many different arrangements or 'words' can be made from the letters in the word 'CONTINUOUS' given that we require:

1 3 letter words, with repetition allowed.

2 3 letter words, without letter repetition.

3 4 letter words, without letter repetition.

4 4 letter words, with exactly one vowel, and no repetition.

5 5 letter words that must begin with T, and repetition is allowed.

6 5 letter words, without repetition, and including exactly 2 vowels.

In how many ways can 8 members of a basketball team be seated:

7 Along a bench?

8 In a circle?

Calculate:

9 ${}^{12}P_5$

10 ${}^{7}P_2 + {}^{11}P_9$

11 ${}^{3}P_3 \times {}^{8}P_1$

There are 3 Australians, 7 Americans, 4 Italians, 6 Japanese and 1 African playing an online game together. At one point in the game, 2 players must form a team.

12 How many teams could be formed?

13 How many teams could be formed where both players are from the same country?

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14 What is the probability that the team consists of players of different nationalities?

At a later point in the online game, the first four players must line up to move to the next level.

15 How many line ups are possible?

16 What is the probability that all 3 Australians are in the line up?

Solutions

A 1440

B 2401

D 840

E 19 958 442

F 210

H 90

I 5040

L 143 640

N 48

O $\frac{1}{1330}$

P 288

R 343

S $\frac{11}{14}$

T 420

V 95 040

Y 40 320

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